Cutoff of the simple exclusion process

with inhomogeneous conductances

A 45-minute blackboard talk

Graduate students and professors in probability. The spoken material is in English.

Running order

Aim to finish the exposition at minute 42. Keep the remaining 3 minutes for questions or interruptions. The 13 main slides are writing prompts, not 13 separate boards to fill. The 4 backup slides are outside the planned 42 minutes.

Time	Slide	Purpose
0–1	1	Opening question
1–5	2	Dynamics, state space and invariant law
5–8	3	Total variation, mixing time and cutoff
8–11	4	The bounded i.i.d. corollary
11–14	5	The full deterministic assumptions
14–17	6	Spectral homogenization input
17–21	7	The slow mode and the factor $1/2$
21–26	8	The lower-bound proof in the bounded case
26–30	9	Height order and coupling
30–34	10	The weighted discrepancy
34–37	11	Exponential drift
37–41	12	Fluctuations, censoring and coalescence
41–42	13	Return to the main theorem
42–45		Discussion or time buffer

Use of the board

Keep a small area visible throughout for

$$r_x = c_x^{-1}, \qquad c_\rho N^\rho \le k \le N/2, \qquad t_{\mathrm{mix}}(\varepsilon) \sim \frac{N^2}{2\pi^2} \log k.$$

Use the remaining space first for the model and definitions, then for the eigenfunction argument, and finally for the height coupling. Draw the particle system, the schematic cutoff curve and the ordered height paths by hand. Each picture should take less than a minute.

If the discussion takes longer than expected

State the two extreme-resistance bounds orally rather than deriving examples. In the lower bound, quote the variance estimate after giving its one-line reason. In the upper bound, keep the definitions of B_t and its drift inequality, then explain the coalescence step in words. Preserve the theorem, the k versus $\sqrt{k}$ calculation and the distinction between drift and coalescence.

Minutes 0–14: the question, model and theorem

Opening sentence. "How long does a system of particles need to forget its initial positions when the rate of exchange varies from bond to bond? I will show that, under suitable resistance assumptions, its leading cutoff time agrees with the homogeneous one."

Write the model. Put k particles on $\{1,\dots,N\}$, at most one per site. A rate-c_x clock at bond $\{x,x+1\}$ swaps the occupations. There are no jumps across the endpoints. Write

$$\Omega_{N,k}=\Big\{\eta\in\{0,1\}^N:\sum_x \eta(x)=k\Big\},\qquad \mathcal{L}f(\eta)=\sum_{x<N} c_x[f(\eta^{x,x+1})-f(\eta)].$$

The uniform measure μ is reversible because the exchange rate is the same in both directions. Thus the inhomogeneity changes the dynamics while preserving the equilibrium law. Assume $k\le N/2$ by particle–hole symmetry.

Write the target. Define

$$d(t)=\max_{\eta}\|P_t^{\eta}-\mu\|_{\mathrm{TV}},\qquad t_{\mathrm{mix}}(\varepsilon)=\inf\{t:d(t)\le\varepsilon\}.$$

Explain total variation as the largest difference in probability of an event. Cutoff at t_N means $t_{\mathrm{mix}}(\varepsilon)/t_N\to 1$ for every fixed $\varepsilon\in(0,1)$. On a schematic graph, time is rescaled by t_N. Avoid suggesting the graph is a proved profile.

State the clean corollary first. Let (r_x) be i.i.d., with $0<a\le r_x\le b$ and $\mathbb{E}r_x=1$. If $c_\rho N^\rho\le k\le N/2$ for some fixed $\rho>0$, then, for almost every environment,

$$\boxed{t_{\mathrm{mix}}(\varepsilon)\sim\frac{N^2}{2\pi^2}\log k}.$$

For example, r_x can be uniform on $[1/2,3/2]$. "Quenched" means that we fix a typical realization of the rates before running the process. Credit Wilson for the homogeneous lower bound and Lacoin for the matching homogeneous cutoff result, as in the manuscript.

Explain the generalization. The actual theorem is deterministic. Allow rates depending on N, suppress that superscript, and assume

$$\frac{1}{N}\max_{2\le m\le N}\Big|\sum_{x<m} r_x-(m-1)\Big|\longrightarrow 0,$$
$$\max_{x<N} r_x\le C\exp((\log N)^v),\quad 0<v<1,$$
$$\min_{x<N} r_x\ge a_N,\quad a_N\to 0,\quad a_N\log N\to\infty.$$

Keep the same polynomial condition on k. In one dimension, resistance sums are the natural macroscopic coordinate. Independence is only a convenient way to obtain the first condition.

Transition. "The mixing-time constant comes from the slowest one-particle mode. Let me first explain that mode and then lift it to the particle system."

Minutes 14–26: the slow mode and the lower bound

Take spectral homogenization as input. Let $g = g_1$ solve $\mathcal{L}_{N,1} g = -\lambda_1 g$, with $g(1) = 1$. The paper restates the earlier spectral result:

$$\lambda_1 \sim \frac{\pi^2}{N^2}, \qquad g(x) \approx \cos\left(\frac{\pi(x-1/2)}{N}\right), \qquad c_x |g(x+1) - g(x)| \le C/N.$$

The approximation is uniform in x, and g is strictly decreasing. Sketch the cosine. Its mean over the sites is zero.

Lift the eigenfunction. Define $F(\eta) = \sum_x \eta(x) g(x)$. Cancellation of exchange terms gives $\mathcal{L}F = -\lambda_1 F$. With all particles initially on the left, denoted by $\wedge$,

$$\mathbb{E}_\wedge F(\eta_t) = e^{-\lambda_1 t} F(\wedge) \asymp k e^{-\lambda_1 t}.$$

For a uniform k-subset, $\mathbb{E}_\mu \eta(x) = k/N$ and, for $x \neq y$, $\mathbb{E}_\mu[\eta(x)\eta(y)] = k(k-1)/(N(N-1))$. Hence

$$\mu(F) = 0, \qquad \mathrm{Var}_\mu F = \frac{k(N-k)}{N(N-1)} \sum_x g(x)^2 \asymp k.$$

The key calculation. Write $k e^{-\lambda_1 t} \approx \sqrt{k}$ and solve for t. Say: "The factor one half appears because the equilibrium fluctuation scale is the square root of the number of particles." This predicts the scale but does not yet prove total-variation mixing.

A complete short lower bound in the bounded case. Define the instantaneous quadratic variation

$$\Gamma F(\eta) = \sum_{x<N} c_x [g(x+1) - g(x)]^2 [\eta(x+1) - \eta(x)]^2.$$

Since $r_x \le b$ and at most $2k$ edges separate unequal occupations, $\Gamma F \le Ck/N^2$. For a deterministic initial configuration, the martingale associated with the eigenfunction gives

$$\mathrm{Var}(F(\eta_t)) = \mathbb{E} \int_0^t e^{-2\lambda_1 (t-u)} \Gamma F(\eta_u)\, du \le Ck.$$

At $t_- = (\log k)/(2\lambda_1) - s/\lambda_1$, the mean m_t is at least $c e^s \sqrt{k}$. For $E_t = \{F \ge m_t/2\}$, Chebyshev gives

$$P^\wedge_{t_-}(E_t^c) + \mu(E_t) \le C e^{-2s}, \qquad d(t_-) \ge 1 - C e^{-2s}.$$

Choose s large depending on the tolerance. This proves the lower bound in the bounded setting. The full resistance-LLN result uses a special initial law for sparse particles; keep that argument for discussion.

Transition. "The slow mode gives a way to distinguish the process from equilibrium. For the upper bound, we have to show that every initial configuration has lost its information."

Minutes 26–34: order, coupling and the weighted area

Write the height function.

$$h^\eta(x) = \sum_{y=1}^{x} \eta(y) - kx/N, \qquad h^\eta(0) = h^\eta(N) = 0.$$

An occupied site contributes slope $1 - k/N$, a vacant site slope $-k/N$. Swapping a neighboring particle and vacancy flips a corner. Draw the upper tent $h^\wedge$ and the lower tent $h^\vee$, with a path between them.

Explain the coupling. The paper couples all height processes with shared clocks indexed by location, height and flip direction. This preserves pointwise height order. It is this corner construction that we use here. Then

$$h_t^\vee \le h_t^\eta \le h_t^\wedge, \qquad h_t^\vee \le h_t^\mu \le h_t^\wedge.$$

If the two extremal processes meet, every process in the coupling has met. Thus, for $T = \inf\{t : h_t^\wedge = h_t^\vee\}$,

$$d(t) \le \mathbb{P}(T > t).$$

It is useful to compare the upper and lower paths separately with the stationary path. Write $T_1 = \inf\{t : h_t^\wedge = h_t^\mu\}$ and define T_2 similarly. Then $T = \max(T_1, T_2)$.

Why the weights come from a slightly larger segment. Put conductance 1 outside the physical interval, extending it by about δN sites on both sides. Let G be the decreasing first eigenfunction on that enlarged interval. The particle system itself still runs on $[1, N]$. Set

$$w_x = G(x) - G(x+1) > 0, \qquad \bar{\lambda}_1 \sim \frac{\pi^2}{(1+2\delta)^2 N^2}.$$

The physical bonds now stay away from the enlarged endpoints. Spectral homogenization gives

$$c_\delta/N \le c_x w_x \le C_\delta/N \qquad (1 \le x < N).$$

This avoids the small flux near the endpoints of the original first mode.

Write the discrepancy.

$$B_t = \sum_{x=1}^{N-1} w_x \big[h_t^\wedge(x) - h_t^\mu(x)\big] \ge 0.$$

Positive weights and order imply $B_t = 0$ exactly when the two paths coincide. The manuscript normalizes this quantity as $A_t = B_t/d_{\min}$, where $d_{\min} = \min_x c_x w_x \asymp_\delta N^{-1}$. Use B_t in the main talk to keep the notation lighter.

Transition. "This discrepancy has a useful negative drift. That drift brings us into the fluctuation range. After that, active corner flips finish the coupling."

Minutes 34–42: drift, fluctuations and the conclusion

Drift calculation. The height generator is

$$(\mathcal{L}h^\eta)(x) = c_x\big[h^\eta(x+1) + h^\eta(x-1) - 2h^\eta(x)\big].$$

Summation by parts against w_x gives $-\bar{\lambda}_1 B$ plus two nonpositive boundary terms. With the generator of the coupled pair,

$$\mathcal{L}^{\mathrm{coup}} B \le -\bar{\lambda}_1 B, \qquad \mathbb{E}B_t \le Cke^{-\bar{\lambda}_1 t}.$$

At $t_a = (1+\delta/2)(\log k)/(2\bar{\lambda}_1)$,

$$\mathbb{E}B_{t_a} \le Ck^{1/2-\delta/4}, \qquad \mathbb{P}(B_{t_a} > k^{1/2-2\psi}) \le Ck^{-\delta/4+2\psi} \to 0$$

when $0 < \psi < \delta/8$. The exponent calculation takes one line. Emphasize that a small mean does not itself imply equality of paths.

Explain the remaining proof in words. The path discrepancy has active corner flips as long as it is nonzero. To quantify their number, the proof bounds the maximal weighted height difference and the length of monotone runs in the stationary path. These yield a lower bound on the quadratic variation of the martingale part of $A_t = B_t/d_{\min}$.

A nonnegative supermartingale cannot accumulate too much quadratic variation without falling to a lower level. The proof applies this through successive levels of the form $d_{\min}^{-1}k^{1/2-i\psi}$. The final descent costs at most order N^2/a_N with high probability, and

$$\frac{N^2/a_N}{N^2\log k} \longrightarrow 0$$

because $\log k$ is comparable to $\log N$ and $a_N \log N \to \infty$.

Name the separate censoring input. A technical estimate for the profile from the maximal configuration comes from the discrete heat equation and censoring. The proof first brings a finite skeleton of heights close to its equilibrium law. It then freezes those skeleton sites and equilibrates the segments between them. Censoring from the maximal state bounds the original dynamics. This is an independent ingredient in the coupling argument, not an assumption that the desired worst-case mixing bound already holds.

Write the outcome, then take limits. The upper and lower extremal comparisons give

$$\mathbb{P}\left(T > \frac{1+\delta}{2\bar{\lambda}_1}\log k\right) \longrightarrow 0.$$

The resulting leading ratio is at most $(1+\delta)(1+2\delta)^2$. First let $N \to \infty$ for fixed δ, then let $\delta \downarrow 0$. Combine with the lower bound.

Closing sentence. "Resistance homogenization fixes the slow relaxation rate. The first mode decays from order k to order $\sqrt{k}$, producing the factor one half. The height coupling shows that no further obstruction remains at this leading time scale." Stop at minute 42.

Discussion notes and source map

Why is mean resistance the normalization? If the mean is $m \in (0, \infty)$, replace r_x by r_x/m and c_x by mc_x. The normalized process runs m times faster. Under the corresponding normalized assumptions, the original mixing time is therefore

$$t_{\mathrm{mix}}(\varepsilon) \sim \frac{mN^2}{2\pi^2} \log k.$$

The effective one-dimensional rate is the reciprocal of mean resistance, not mean conductance.

What changes for sparse particles in the full theorem? Under the resistance LLN alone, a crude bracket estimate is $O(N)$ rather than $O(k)$. For $k < N/64$, the paper samples $2k$ equilibrium particles and retains the leftmost k. The resulting initial law ν has $\mathbb{E}_\nu F \geq ck$, and a domination coupling gives $\mathbb{E}_\nu \eta_t(x) \leq 2k/N$. This recovers an $O(k)$ martingale variance. Appendix A controls the random initial contribution at the chosen time. Convexity of total variation makes any mixture of initial states admissible for a worst-case lower bound. The lower-bound proposition only requires $k \geq (\log N)^{1+\gamma}$; the full cutoff theorem has the stronger polynomial condition.

What is the bracket inequality? With

$$H_t = \max_{x<N} w_x(h_t^{\wedge}(x) - h_t^{\mu}(x))$$

and Q_t the longest monotone run of h_t^{μ}, the paper proves, before T_1,

$$\frac{d}{dt}\langle A\rangle_t \geq a_N \max\left\{1, \frac{A_t d_{\min}}{3H_t Q_t}\right\}.$$

Here $\langle A\rangle$ denotes the predictable bracket of the martingale part of A. This is a technical backup, not a formula to develop in the main 45 minutes.

What does the theorem leave outside its scope? The theorem does not cover fixed k or every integrable resistance law. It gives the leading cutoff time, without a sharp window or a limiting profile. For $k > N/2$, replace k by the number of vacancies using particle–hole symmetry, subject to the same growth assumptions. The extension to a circle is a direction mentioned in the manuscript, rather than a current-literature claim in this talk.

Where each part comes from in the attached manuscript

Talk material	Manuscript location
Model and full theorem	Section “Model and result”, resistance LLN, three assumptions, main mixing theorem
Spectral input	Theorem A, eigenvalue, eigenfunction and weighted-derivative convergence
Lower bound	Section “The lower bound on the mixing time” and covariance appendix
Height order and coupling	Section “Preliminaries”, partial order and graphical construction
Weighted area and descent	Section “The upper bound on the mixing time”
Censoring input	Appendices “Height function” and proof of the maximal-state proposition